

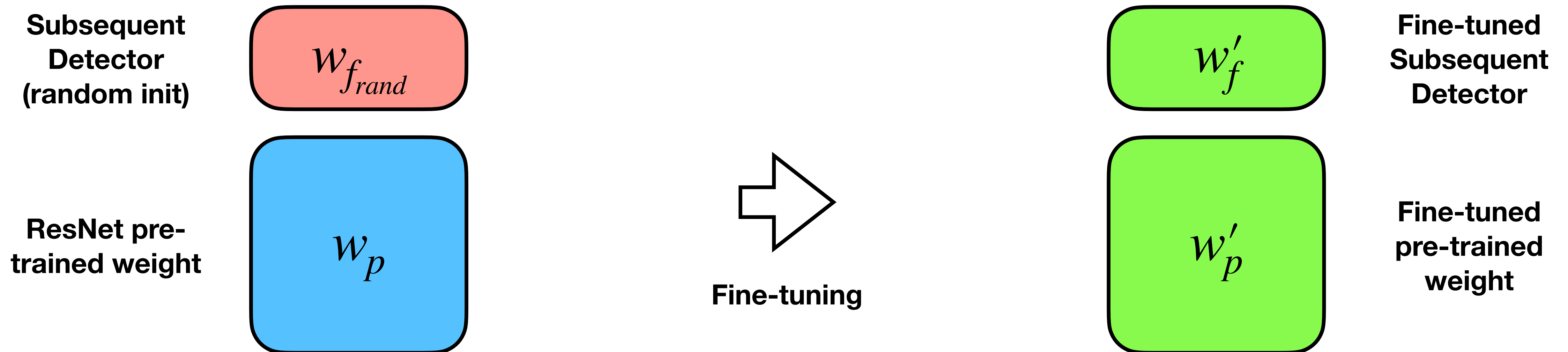
이미지처리팀

Proper Reuse of Image Classification Features Improves Object Detection

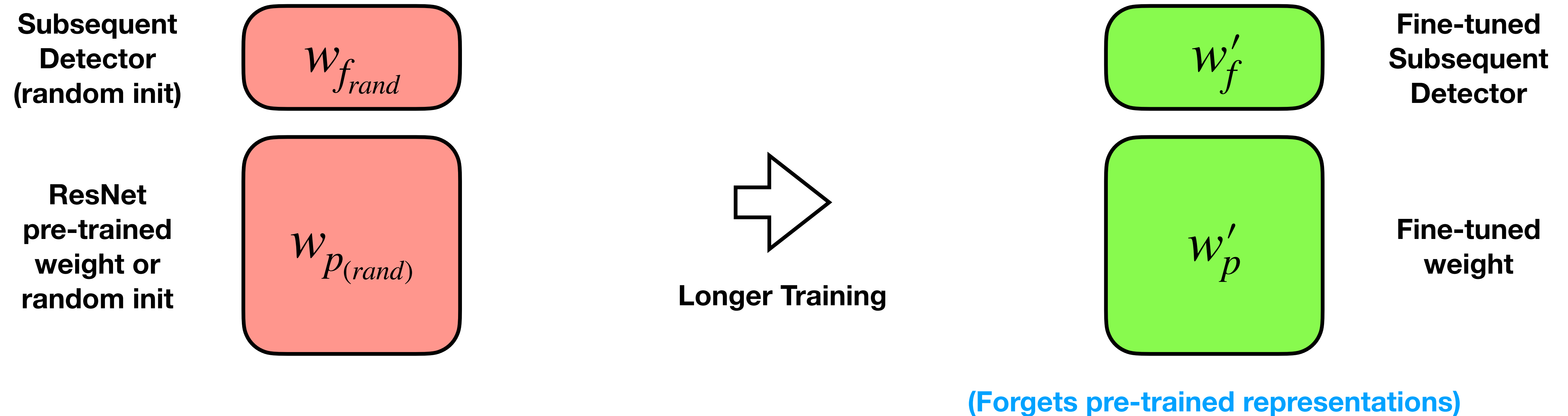
CVPR22 Paper

김병현, 김선옥, 김현진, 안종식, 이찬혁, 홍은기, **현청천**

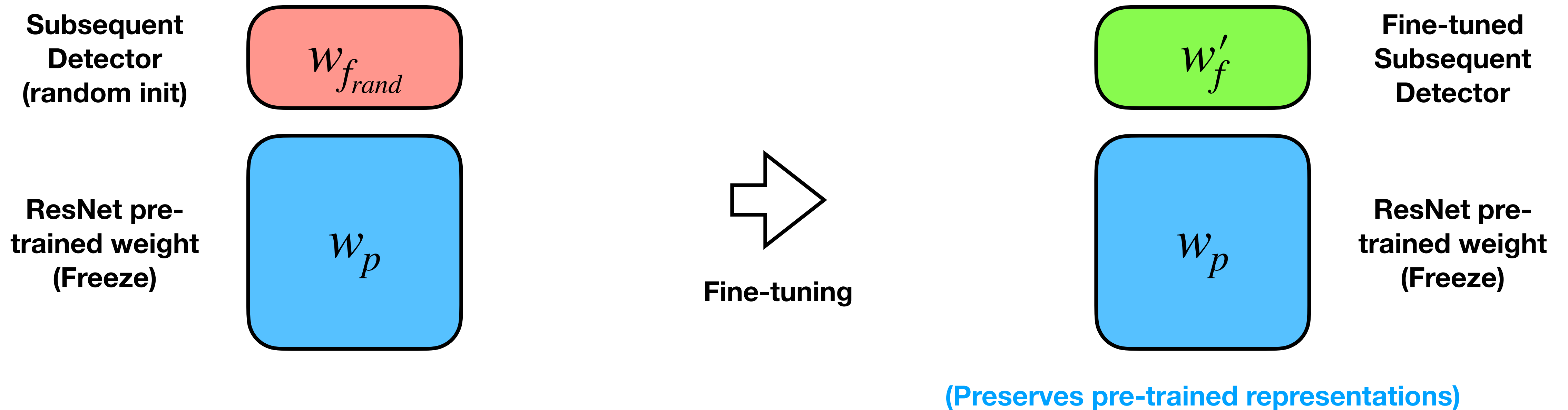
2022.10.29



일반적으로 가장 많이 사용되는 방법으로 성능이 가장 좋은 것으로 알려져 있음



도메인 관련 학습할 데이터가 많은 경우 fine-tuning과 비슷한 성능 달성



이 논문이 제안하는 학습방법으로 비교적 적은 학습 리소스 (FLOP, MEM) 사용으로 좋은 성능 달성

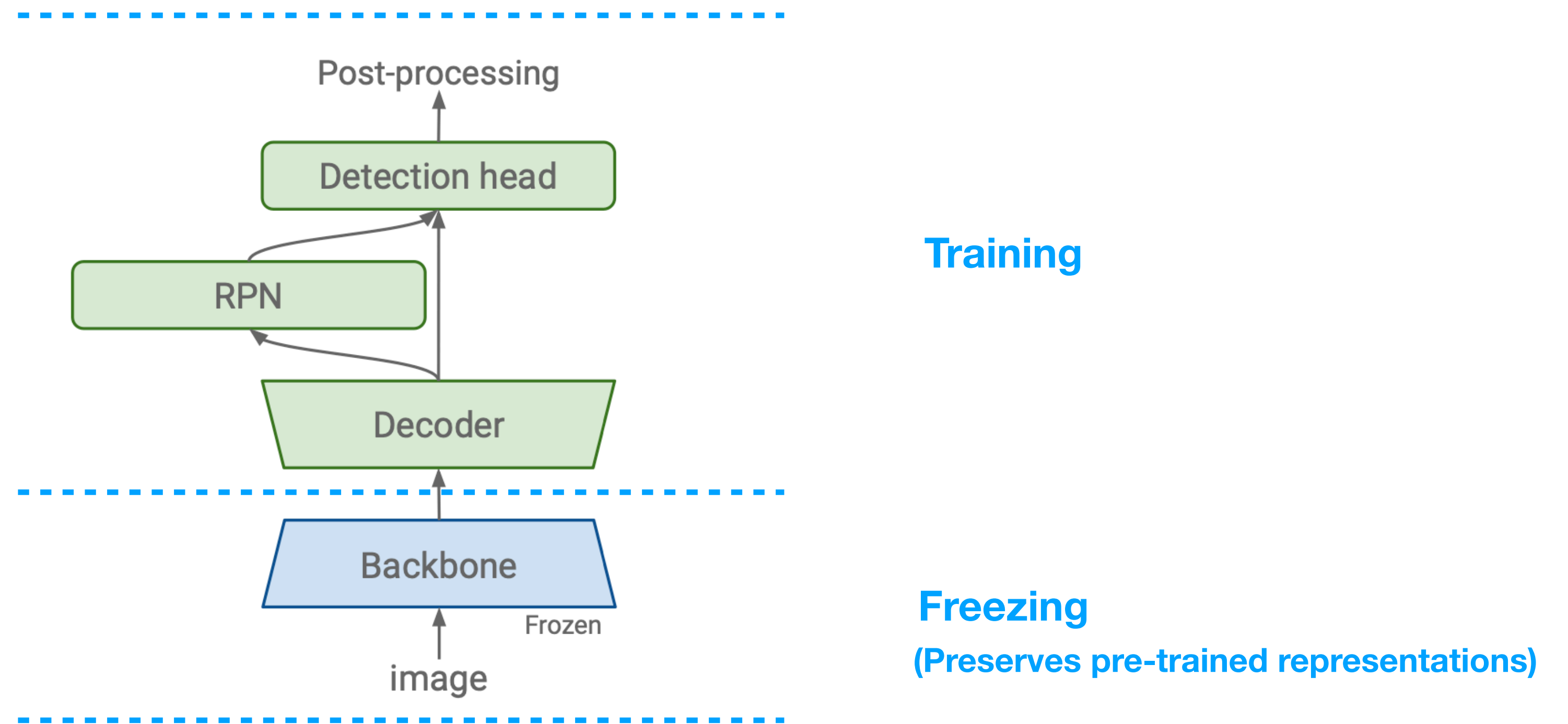


Figure 2. A sketch of the Faster-RCNN detection model that serves as an example. Our proposed training procedure freezes the backbone after initializing it from a classification task and trains the other components from scratch.

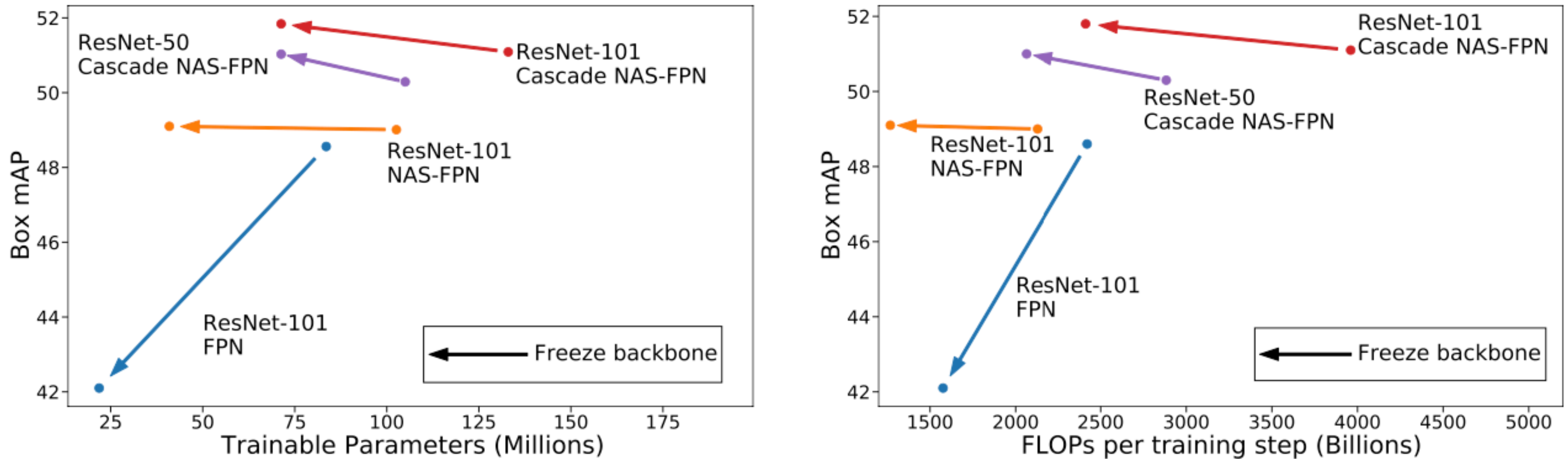


Figure 1. Impact on object detection performance of preserving and not fine-tuning features learned on ImageNet. The arrows indicate impact on each model, when trained with frozen backbone weights. As long as the remaining detector components have enough capacity (**left**), freezing increases performance while significantly reducing resources used during training (**right**).

Subsequent Detector의 크기와 구조가 성능향상에 중요한 역할
Long-tail과 같은 어려운 문제에서도 좋은 성능 달성

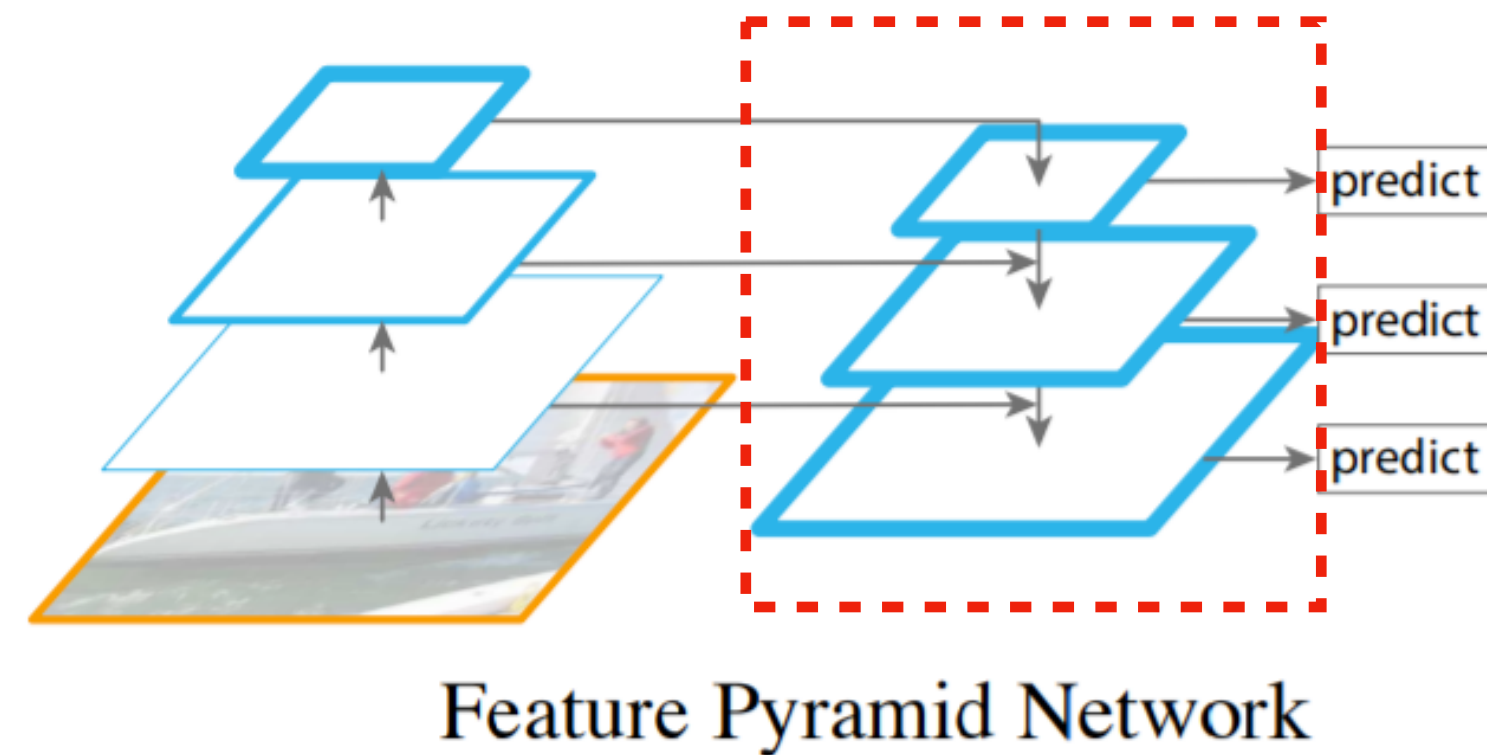
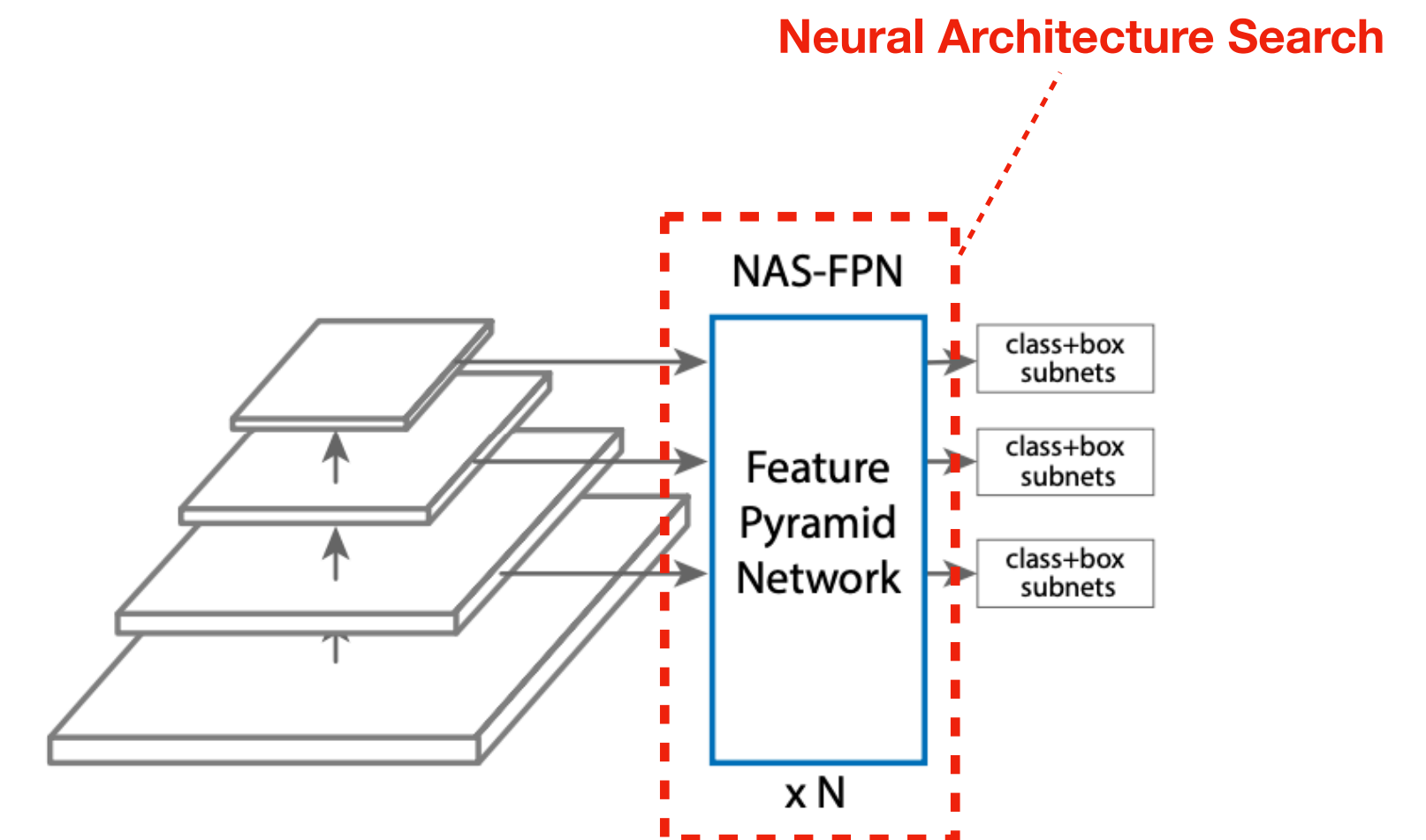
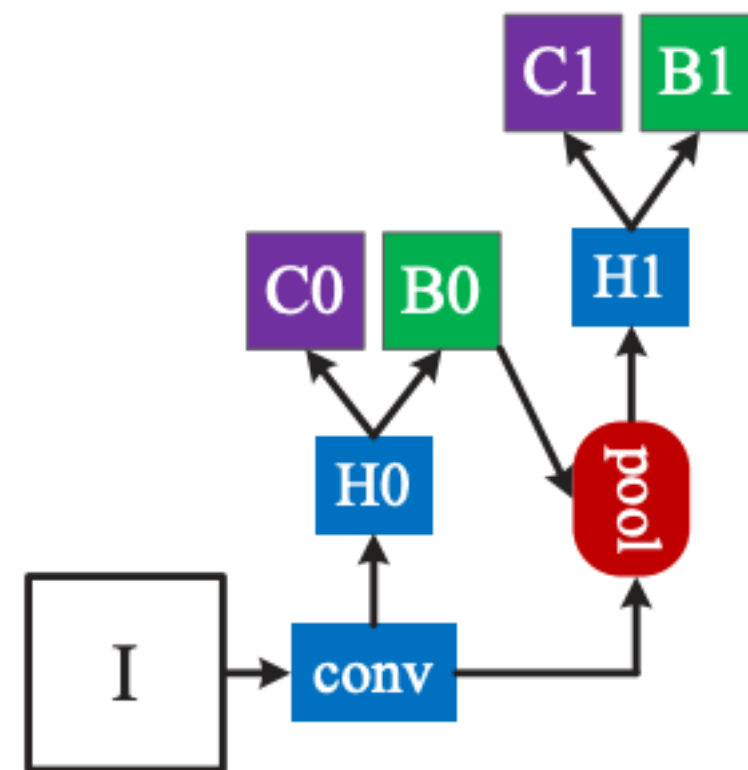
FPNNAS-FPN

Figure 2: RetinaNet with NAS-FPN. In our proposal, feature pyramid network is to be searched by a neural architecture search algorithm. The backbone model and the subnets for class and box predictions follow the original design in RetinaNet [23]. The architecture of FPN can be stacked N times for better accuracy.

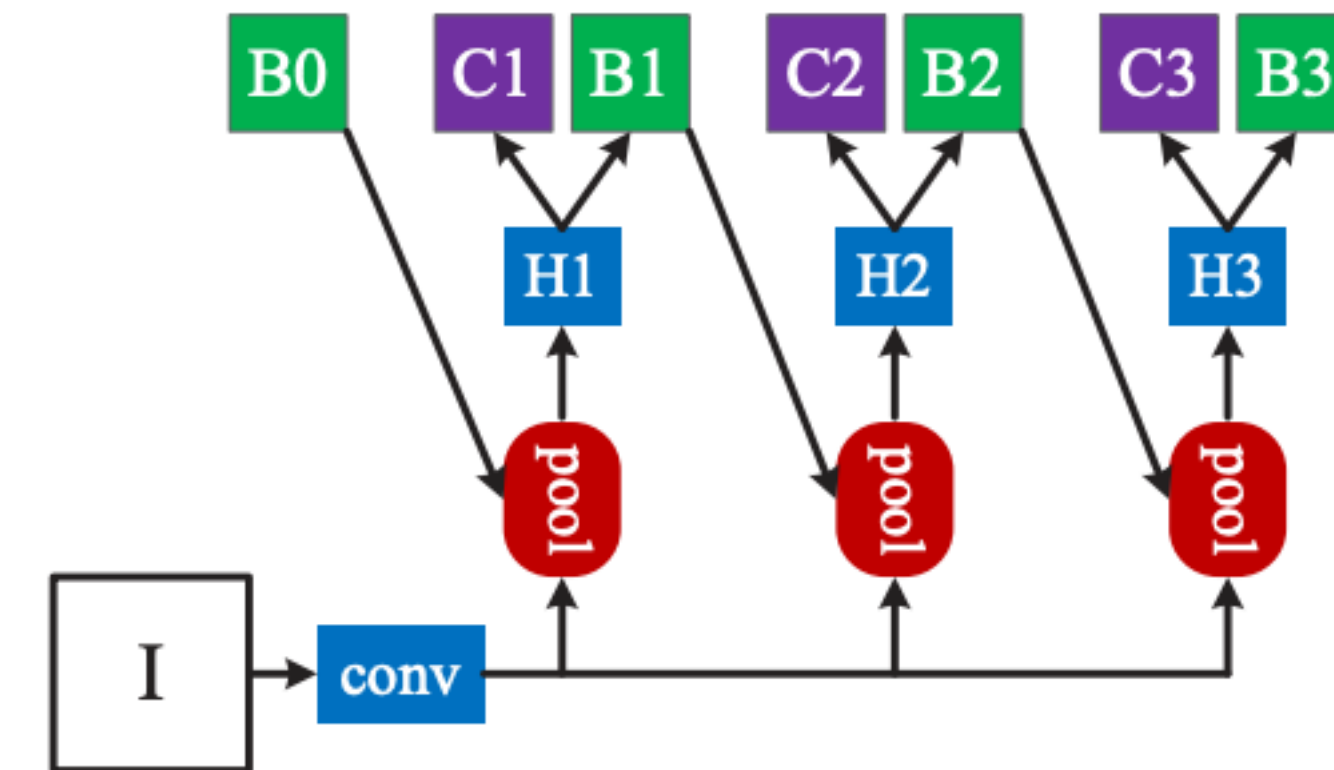
NAS-FPN is a Feature Pyramid Network that is discovered via **Neural Architecture Search** in a novel scalable search space covering all cross-scale connections.

Faster R-CNN



Faster R-CNN

Cascade R-CNN



Cascade R-CNN

Cascade R-CNN is a multi-stage extension of the R-CNN.

ImageNet



Figure 1: A snapshot of two root-to-leaf branches of ImageNet: the **top** row is from the mammal subtree; the **bottom** row is from the vehicle subtree. For each synset, 9 randomly sampled images are presented.

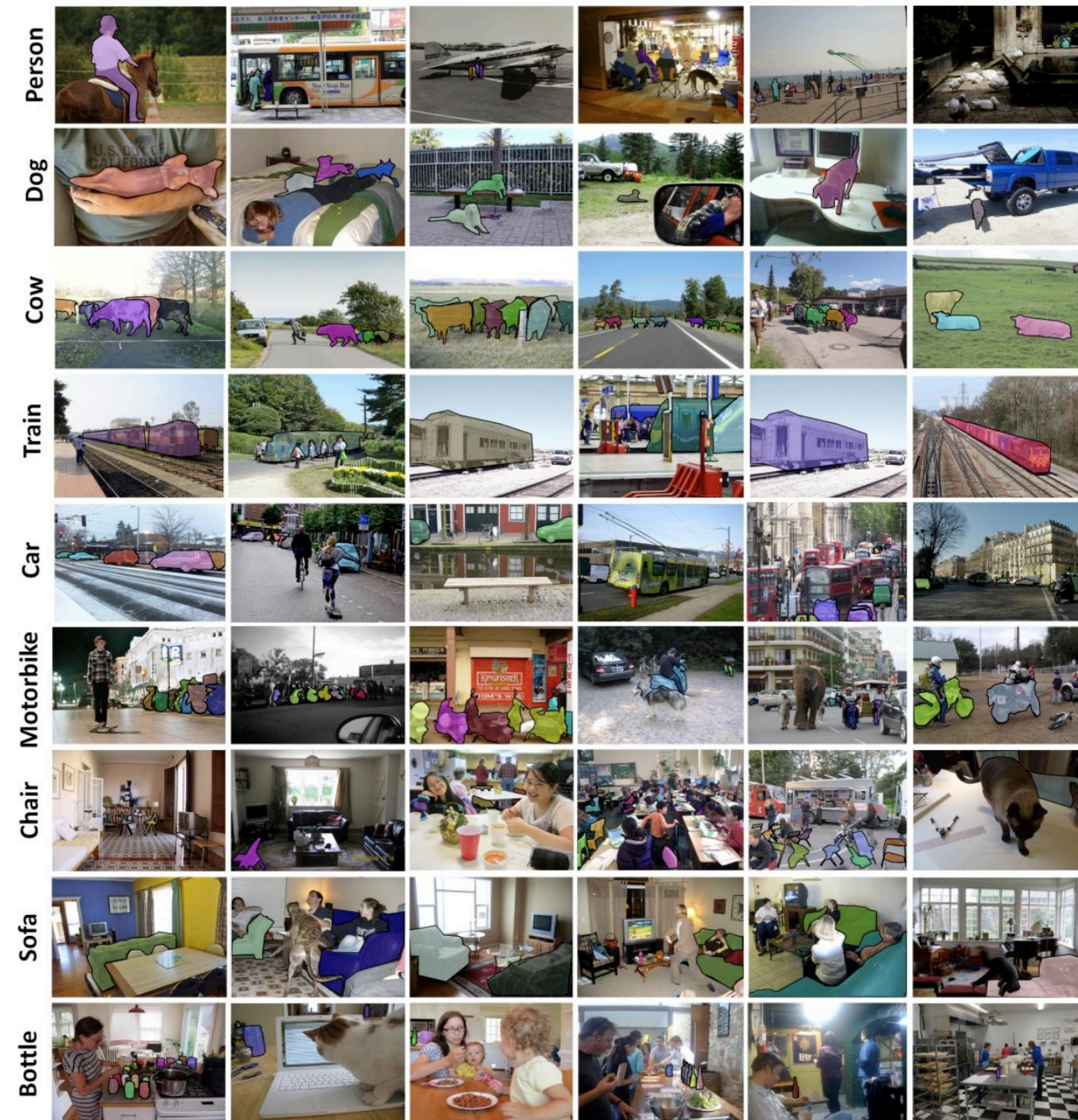
- 1.2M images
- 1,000 classes

JFT-300M



- 300M images
- 18,291 classes
- Labeled using an algorithm
- 20% label error

MSCOCO



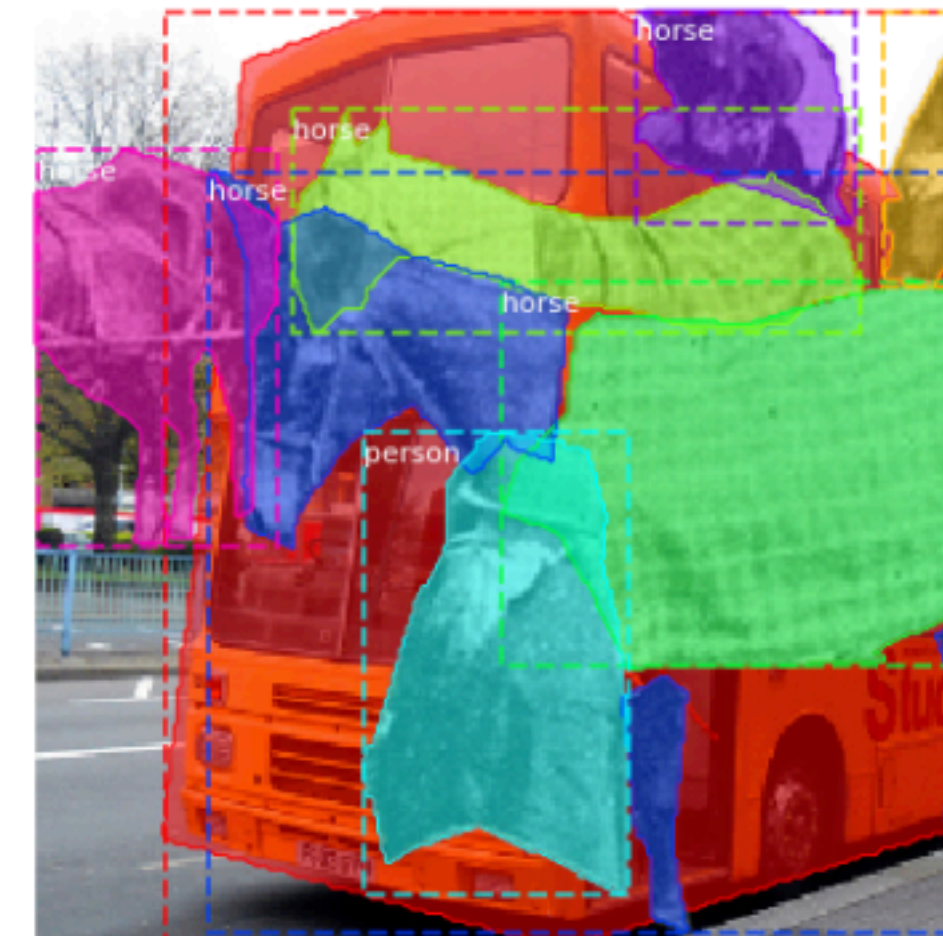
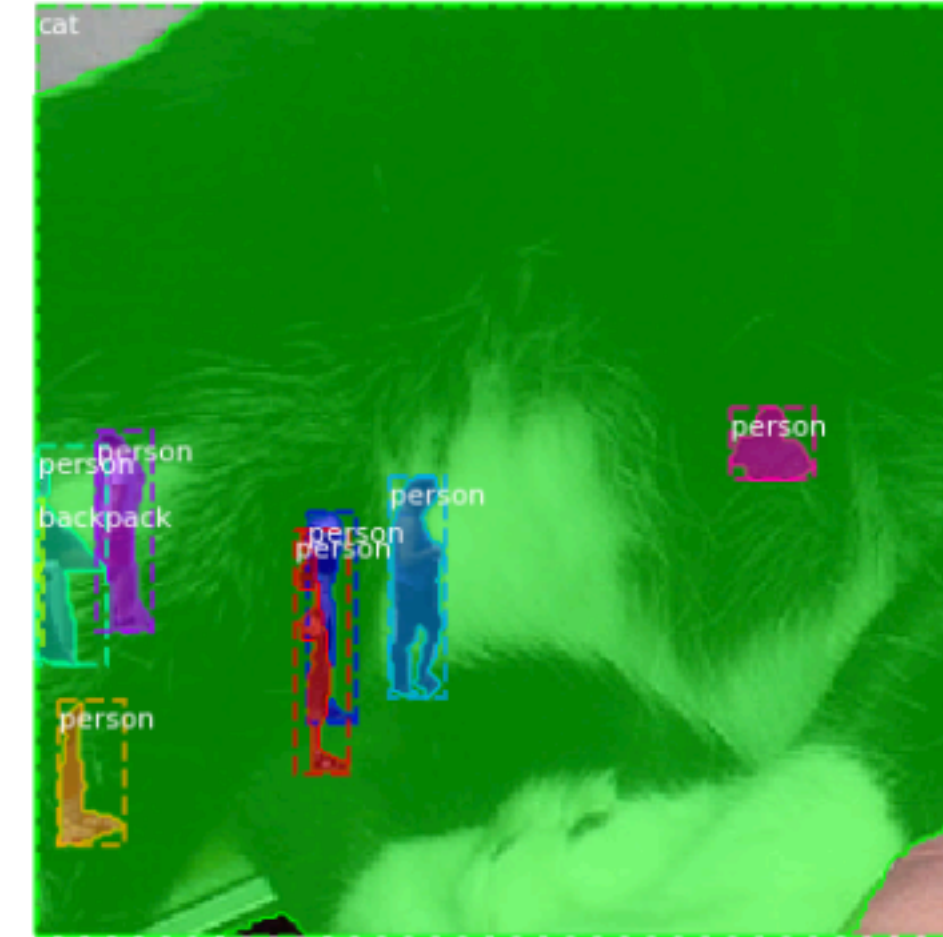
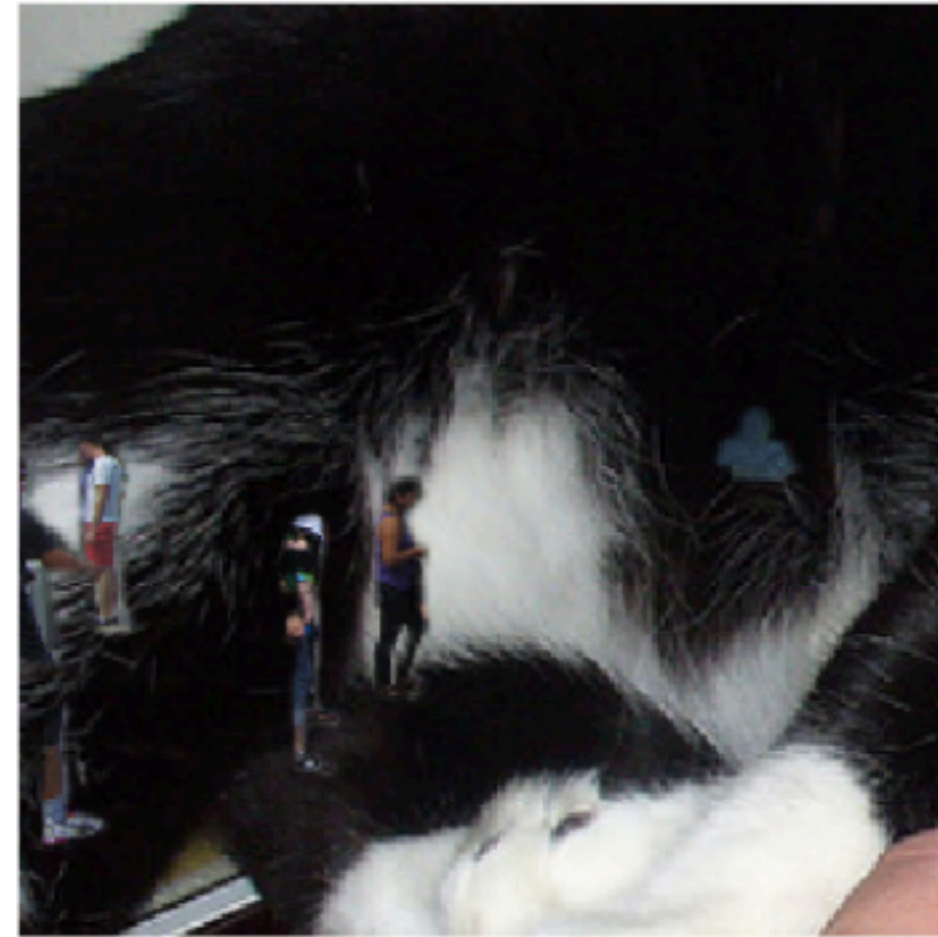
- 118K images
- 91 classes

LVIS 1.0

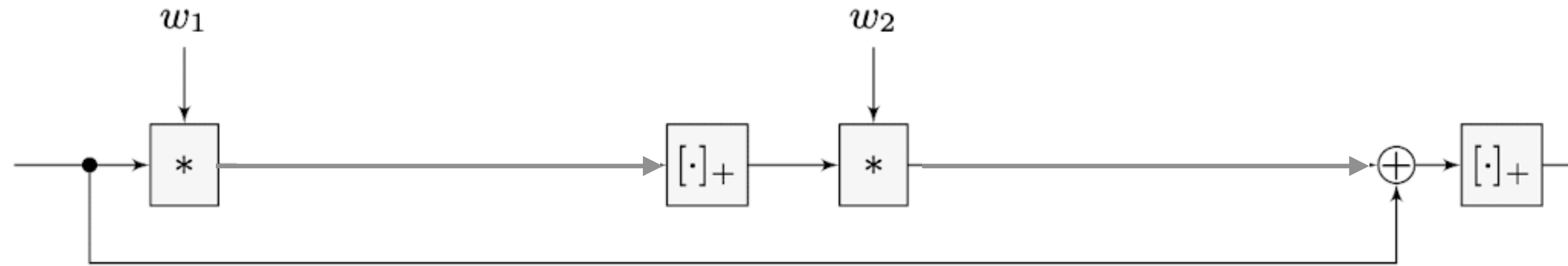


- 100K images
- 1,203 classes
- Long tail distribution

Copy-Paste



ResNet Residual Block



Residual Adapter Block

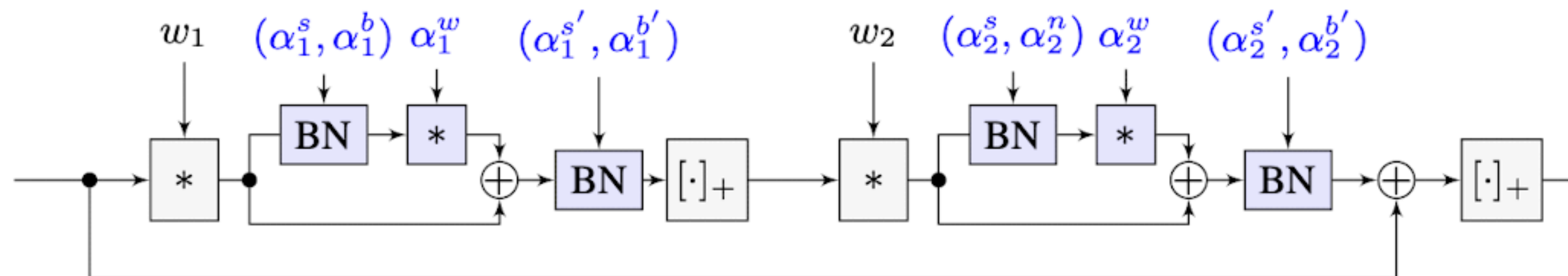


Figure 2: **Residual adapter modules.** The figure shows a standard residual module with the inclusion of adapter modules (in blue). The filter coefficients (w_1, w_2) are domain-agnostic and contains the vast majority of the model parameters; (α_1, α_2) contain instead a small number of domain-specific parameters.

- **Freeze:** w_1, w_2
- **Fine-Tune:** $(\alpha_1^s, \alpha_1^b), \alpha_1^w, (\alpha_1^{s'}, \alpha_1^{b'}), (\alpha_2^s, \alpha_2^b), \alpha_2^w, (\alpha_2^{s'}, \alpha_2^{b'})$

• 72 epoch vs 600 epoch

Model	Pretraining	mAP	AP @ 50
NAS-FPN + ResNet-101 -----	ImageNet	44.0	63.0
	+ Freeze backbone	(+0.0) 44.0	(+0.2) 63.2
	JFT-300M	44.8	63.7
	+ Freeze backbone	(+0.8) 45.6	(+1.5) 65.3
FPN + ResNet-101 ----	ImageNet	42.6	62.2
	+ Freeze backbone	(-7.3) 35.3	(-6.0) 56.2
	JFT-300M	43.7	63.0
	+ Freeze backbone	(-8.4) 35.3	(-4.8) 58.2

Table 13. One-stage detector performance (RetinaNet, 72 epochs). Similarly to the two-stage detectors, models with frozen features based on NAS-FPN obtain similar performance e significantly reducing resources used during training.

Model	Pretraining	mAP	AP @ 50
NAS-FPN + ResNet-101	From scratch	42.6	61.7
	ImageNet	44.3	63.3
	+ Freeze backbone	(+1.5) 45.8	(+1.9) 65.2
	JFT-300M	44.0	62.8
	+ Freeze backbone	(+2.4) 46.4	(+3.5) 66.3
FPN + ResNet-101	From scratch	43.5	63.2
	ImageNet	43.8	63.6
	+ Freeze backbone	(-7.6) 36.2	(-6.2) 57.4
	JFT-300M	44.3	63.7
	+ Freeze backbone	(-8.3) 36.0	(-4.6) 59.1

Table 14. One-stage detector performance (RetinaNet) under longer training (600 epochs) shows similar trends to the two-stage detectors results: models with Nas-FPN benefit from feature preservation. Training for longer improved the results and relative gain of feature preservation for models based on NAS-FPN but did not help closing the performance gap on those based on FPN.

- 72 epochs
- NAS-FPN: backbone을 freeze하면 성능 향상
- FPN: backbone을 freeze하면 성능 감소

- 600 epochs
- NAS-FPN
 - backbone을 freeze하면 성능 향상
 - 72 epoch에 비해서 추가 성능 향상
- FPN: backbone을 freeze하면 성능 감소

Backbon freeze시 많은 epochs 학습이 성능 향상

- The relative benefit from pre-training on larger classification datasets is clear when freezing the backbone.

Model	Pretraining	mAP	AP @ 50
ResNet-101 + FPN	From scratch	48.4	70.1
	ImageNet	48.6	70.5
	+ Freeze backbone	(−6.5) 42.1	(−5.2) 65.3
	JFT-300M	48.7	70.5
	+ Freeze backbone	(−5.6) 43.1	(−3.3) 67.2
ResNet-101 + NAS-FPN	From scratch	47.2	68.2
	ImageNet	49.0	70.0
	+ Freeze backbone	(+0.1) 49.1	(+0.3) 70.3
	JFT-300M	49.1	70.2
	+ Freeze backbone	(+1.0) 50.1	(+1.5) 71.8
ResNet-50 + NAS-FPN + Cascade	From scratch	50.0	68.1
	ImageNet	50.3	68.0
	+ Freeze backbone	(+0.7) 51.0	(+1.1) 69.1
	JFT-300M	50.4	68.1
	+ Freeze backbone	(+1.7) 52.1	(+2.3) 70.4
ResNet-101 + NAS-FPN + Cascade	From scratch	50.4	68.4
	ImageNet	51.1	69.1
	+ Freeze backbone	(+0.8) 51.8	(0.8) 69.9
	JFT-300M	51.1	69.0
	+ Freeze backbone	(+1.7) 52.8	(+2.1) 71.1

Table 1. With a powerful-enough detector, freezing the backbone to its pre-trained initialization during detection training outperforms fine-tuning the backbone or training it from scratch.

- No freeze backbone
 - ImageNet vs JFT-300 비슷한 성능
- Freeze backbone
 - ImageNet 보다 JFT-300가 최대 0.9 mAP의 성능 향상

Backbone freeze시 많은 데이터로 pre-train이 성능 향상

- The relative benefit from pre-training on larger classification datasets is clear when freezing the backbone.

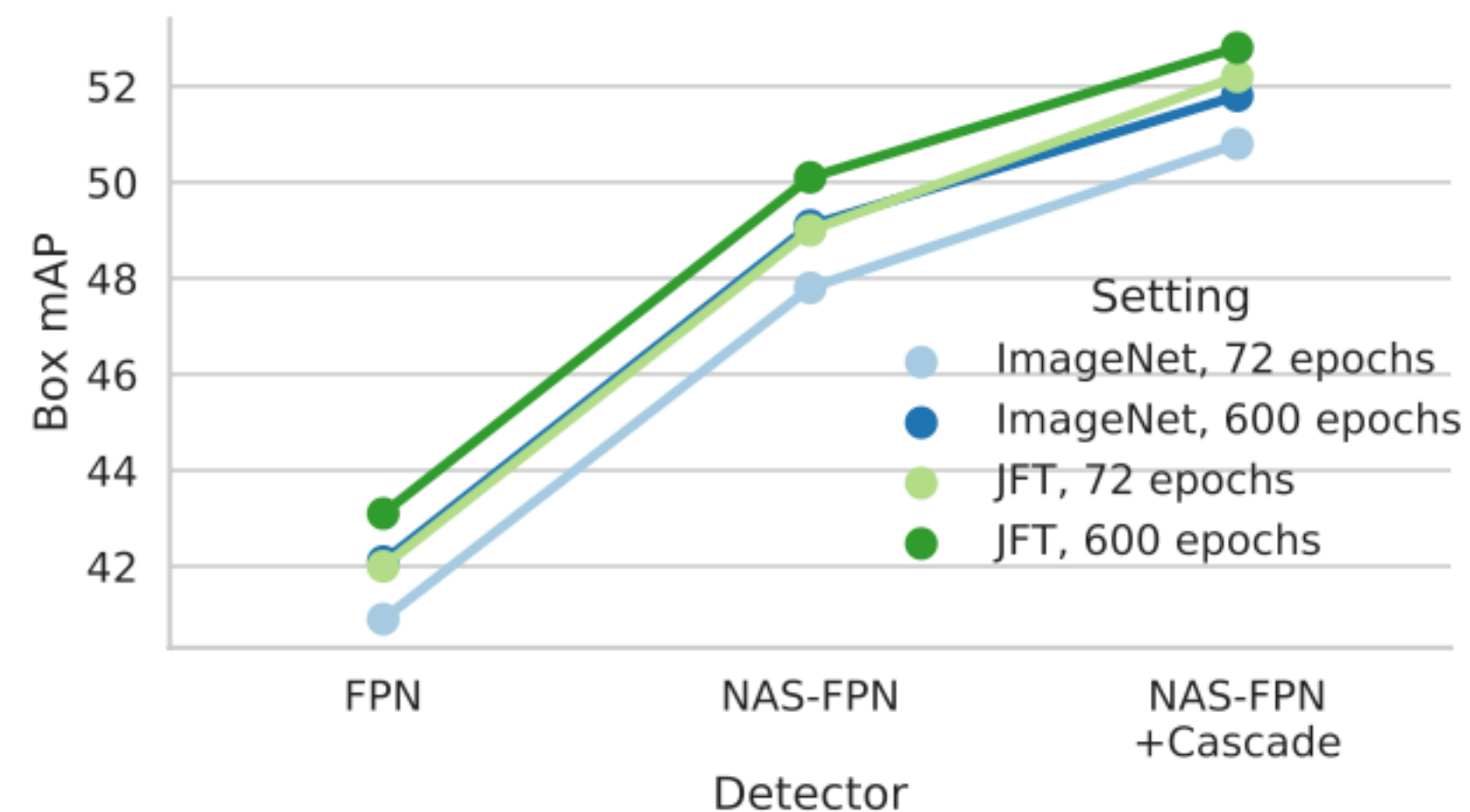


Figure 6. Performance of different detectors preserving the same set of weights (ResNet-101). As the backbones are frozen, it is possible to disentangle the relative benefit from pre-training on ImageNet *versus* JFT-300M from other confounding factors. Pre-training on a larger image classification dataset (JFT-300M) has a clear (and similar) benefit across detectors with different compositions (FPN, NAS-FPN, NAS-FPN + Cascade). Training longer benefits all variations.

- Backbone Freeze시 아래 순서와 같이 성능이 향상
 - JFT 600 epochs
 - JFT 72 epochs
 - ImageNet 600 epochs
 - ImageNet 72 epochs

- Proper reuse of image classification features improves performance.

Paper	Pre-training	Schedule	Freeze?	mAP
<i>Pre-training on larger classification datasets helps.</i>				
Sun et al. [47]	ImageNet	Short	Yes	47.8
	JFT	Short	Yes	49.0
<i>Pre-training does not help with longer training schedules.</i>				
He et al. [14]	ImageNet	Long	No	49.0
	JFT	Long	No	49.1
<i>Backbone freezing & high-capacity detector components help.</i>				
Ours	JFT	Long	Yes	50.1

Table 2. We revisit the conclusions from [14, 47] under modern training regimes and best practices. We use their main conclusions and re-train these models for a fair comparison. All results are reported on a ResNet-101 backbone with a NAS-FPN. Note that Sun et al. [47] did not run experiments with NAS-FPN.

- 데이터 수 JFT > ImageNet
- JFT가 ImageNet보다 Freeze(O), Short 학습에서 좋은 성능
- JFT와 ImageNet모두 Freeze(X), Long 학습에서 비슷한 성능
- High capacity detector, JFT, Freeze(O), Long 학습에서 가장 좋은 성능

- Proper reuse of image classification features improves performance.

Model	Pretraining	mAP	AP @ 50
NAS-FPN +ResNet-50	ImageNet	47.0	68.0
	+ Freeze backbone	(+0.1) 47.1	(+0.3) 68.3
	JFT-300M	47.5	68.9
	+ Freeze backbone	(+0.4) 47.9	(+0.7) 69.6
ResNet-101	ImageNet	48.2	69.6
	+ Freeze backbone	(-0.4) 47.8	(-0.3) 69.3
	JFT-300M	48.5	69.2
	+ Freeze backbone	(+0.5) 49.0	(+1.3) 70.5
NAS-FPN + Cascade +ResNet-50	ImageNet	49.4	66.8
	+ Freeze backbone	(+0.5) 49.9	(+0.3) 67.1
	JFT-300M	49.9	67.6
	+ Freeze backbone	(+1.1) 51.0	(+1.2) 68.8
+ResNet-101	ImageNet	51.1	68.7
	+ Freeze backbone	(-0.3) 50.8	(-0.3) 68.4
	JFT-300M	50.9	68.5
	+ Freeze backbone	(+1.3) 52.2	(+1.5) 70.0

Table 12. Impact on performance from frozen representation associated with the use of NAS-FPN under shorter schedule regime (72 epochs). Freezing from ImageNet takes longer to converge as the increase in accuracy observed on NAS-FPN and NAS-FPN +Cascade models is smaller than the observed in longer training regimes. Models composed with NAS-FPN and frozen backbone produce matching or superior performance while consuming fewer resources during training.

- Short (72 epoch) 학습에서의 성능비교
 - 큰 decoder (NAS-FPN + Cascade)에서 가장 많은 성능 향상

Backbone freeze시 큰 decoder에서 성능향상

- Feature preservation benefits are bounded by the remaining trainable capacity.

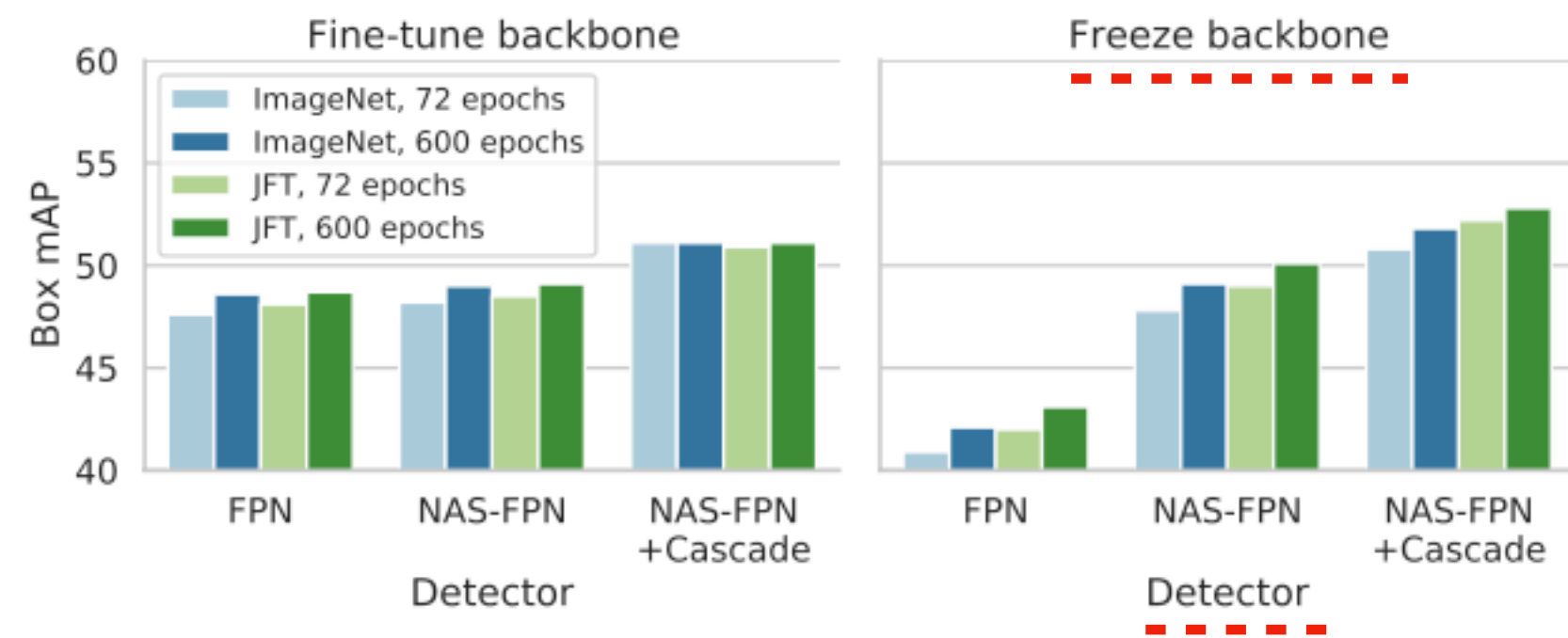


Figure 3. Detectors using a ResNet-101 backbone. Sufficient capacity on components trained is important in realizing the benefits of preserving the representation from a large classification dataset.

- Freeze시 아래 항목에 따른 성능변화
 - decoder의 크기
 - pre-train data량
 - 학습 epochs 횟수

Backbone freeze시 decoder 크기, pre-train data 량, epochs 수 등의 영향을 받음

- Feature preservation benefits are bounded by the remaining trainable capacity.

Detection model	#Params (Million)		Δ mAP
	Original	Trained	
FPN	83.5	(26.1%) 21.9	−6.5
NAS-FPN	102.6	(39.9%) 40.9	+0.1
Cascade + NAS-FPN	132.9	(53.6%) 71.3	+0.7

Table 3. Number of parameters that are trained when keeping the backbone frozen (a ResNet-101) with various detection models. We report Δ mAP as the performance gap from fine-tuned counterparts. Weights initialized from an ImageNet pre-trained model.

- Backbone Freeze시 Decoder 크기에 따른 성능 비교
 - Decoder의 용량 증가에 따른 성능 향상

Backbone freeze시 decoder가 클수록 성능 향상

- Feature preservation benefits are bounded by the remaining trainable capacity.

Pretraining	conv. layers	mAP	mAP @ 50
From scratch	1	48.9	70.7
	2	(−0.5) 48.4	(−0.6) 70.1
– Fine-tune backbone			
ImageNet	1	48.4	69.7
	2	(+0.2) 48.6	(+0.8) 70.5
JFT-300M	1	48.6	70.1
	2	(+0.1) 48.7	(+0.4) 70.5
– With frozen backbone			
ImageNet	1	41.3	64.4
	2	(+0.8) 42.1	(+0.9) 65.3
JFT-300M	1	42.2	66.4
	2	(+0.9) 43.1	(+0.8) 67.2

Table 9. Impact on FPN based models performance from decreasing their RPN capacity by reducing its convolutional layers from two (baseline value) to one layer. Results taken using a Resnet-101 backbone (training for longer). Training from scratch performance is increased with the capacity reduction while the opposite happens for models with pre-trained initialization. Models with frozen backbone have the largest decrease in performance from reducing RPN’s capacity.

- Conv 1개 vs Conv 2개 성능비교
 - Scratch training시 Conv 2가 성능 감소
 - Fine-tuning시 Conv 2가 성능 향상
 - Freeze backbone에서 Conv 2가 성능 향상

Decoder가 클수록 성능이 향상되는 경향이 있음

- Freezing EfficientNet-B7 backbones improves performance on both MSCOCO and LVIS datasets.

MSCOCO (val)	Box mAP	Mask mAP
Swin Transformer [31]*	57.1	49.5
Cascade Eff-B7 NAS-FPN (1280) [12]	54.8	46.9
+ freeze backbone (ours)	(+0.9) 55.7	(+0.7) 47.6
+ Copy-Paste	55.9	47.2
+ freeze backbone (ours)	(+1.1) 57.0	(+1.5) 48.7
+ Self-training Copy-Paste	57.0	48.9
Soft Teacher + Swin-L [52]	59.1	51.0

LVIS (val)	Box mAP	Mask mAP
cRT (ResNeXt-101-32×8d) [22]	—	27.2
LVIS Challenge 2020 Winner [48]†	41.1	38.8
Eff-B7 NAS-FPN (1280) [12]	37.2	34.7
+ freeze backbone (ours)	(+2.2) 39.4	(+2.5) 37.2
+ Copy-Paste	41.6	38.1
+ freeze backbone (ours)	(+1.5) 43.1	(+1.8) 39.9

Table 4. Freezing the backbone improves two strong baselines on MSCOCO and LVIS. when training only on the train2017 split. “Self-training” and [52] use the extra training data. *ImageNet-22K pre-training. †No test-time augmentation. All results are reported on the validation sets.

- EfficientNet-B7 MSCOCO vs LVIS
 - NAS-FPN Decoder를 사용시 freeze backbone에서 성능향상
 - Copy-Paste augmentation에 freeze backbone에서 추가 성능 향상

- When the backbone is frozen, strong data augmentations improves the remaining detector components even further.

LVIS	Box				Mask			
	mAP	mAP _r	mAP _c	mAP _f	mAP	mAP _r	mAP _c	mAP _f
First stage results: regular training								
Copy and Paste [12]	38.5	19.3	37.3	48.2	35.0	19.5	34.9	42.1
+ Freeze backbone	(+1.2) 39.7	(+2.9) 22.2	(+1.5) 38.8	(+0.1) 48.3	(+1.5) 36.5	(+2.9) 22.4	(+1.9) 36.8	(+0.4) 42.5
Second stage: tunes detection-classifier final layer using class-balanced loss								
Copy and Paste [12]	41.6	31.5	39.8	48.0	38.1	32.1	37.1	41.9
+ Freeze backbone	(+1.5) 43.1	(+1.7) 33.2	(+2.1) 41.9	(+0.7) 48.7	(+1.1) 39.9	(+1.5) 33.6	(+2.5) 39.6	(+1.0) 42.9

Table 5. Performance using EfficientNet-B7 + NAS-FPN. Freezing the backbone has the strongest positive performance impact on rare (mAP_r) and common (mAP_c) classes, while still improving frequent (mAP_f) classes. Original first phase results are provided by the authors of [12]. Results without Copy-Paste augmentations can be found in Appendix D.

- Fine-tuning + Augmentation + class-balanced loss
 - Rare class (mAP_r): 성능 향상
 - Common class (mAP_c): 성능 향상
 - Frequent class (mAP_f): 성능 감소
- Backbone freeze + Augmentation + class-balanced loss
 - Rare class (mAP_r): 성능 향상
 - Common class (mAP_c): 성능 향상
 - Frequent class (mAP_f): 성능 향상

- When the backbone is frozen, strong data augmentations improves the remaining detector components even further.

LVIS	Box				Mask			
	mAP	mAP _s	mAP _m	mAP _l	mAP	mAP _s	mAP _m	mAP _l
First stage results: regular training								
Copy and Paste [12]	38.5	30.5	48.2	55.5	35.0	25.7	45.6	53.2
+ frozen backbone	(+1.2) 39.7	(+0.2) 30.7	(+1.4) 49.6	(+2.3) 57.8	(+1.5) 36.5	(+0.3) 26.0	(+1.6) 47.2	(+2.8) 56.0
Second stage: tunes detection-classifier final layer using class-balanced loss								
Copy and Paste [12]	41.6	33.5	51.5	58.1	38.1	28.4	49.0	55.6
+ frozen backbone	(+1.5) 43.1	(+0.6) 34.1	(+1.8) 53.3	(+2.3) 60.4	(+1.8) 39.9	(+0.7) 29.1	(+2.1) 51.1	(+2.7) 58.3

Table 6. Performance using EfficientNet-B7 + NAS-FPN. Freezing the backbone has the strongest positive performance impact on large objects (mAP_l), then medium-sized objects (mAP_m), and finally small objects (mAP_s). Original first phase results are provided by the authors of [12]. Results without Copy-Paste augmentations can be found in Appendix D.

- Backbone freeze + Augmentation + class-balanced loss
 - Small image (mAP_s): 성능 향상
 - Medium image (mAP_m): 성능 향상
 - Large image (mAP_l): 가장 큰 성능 향상

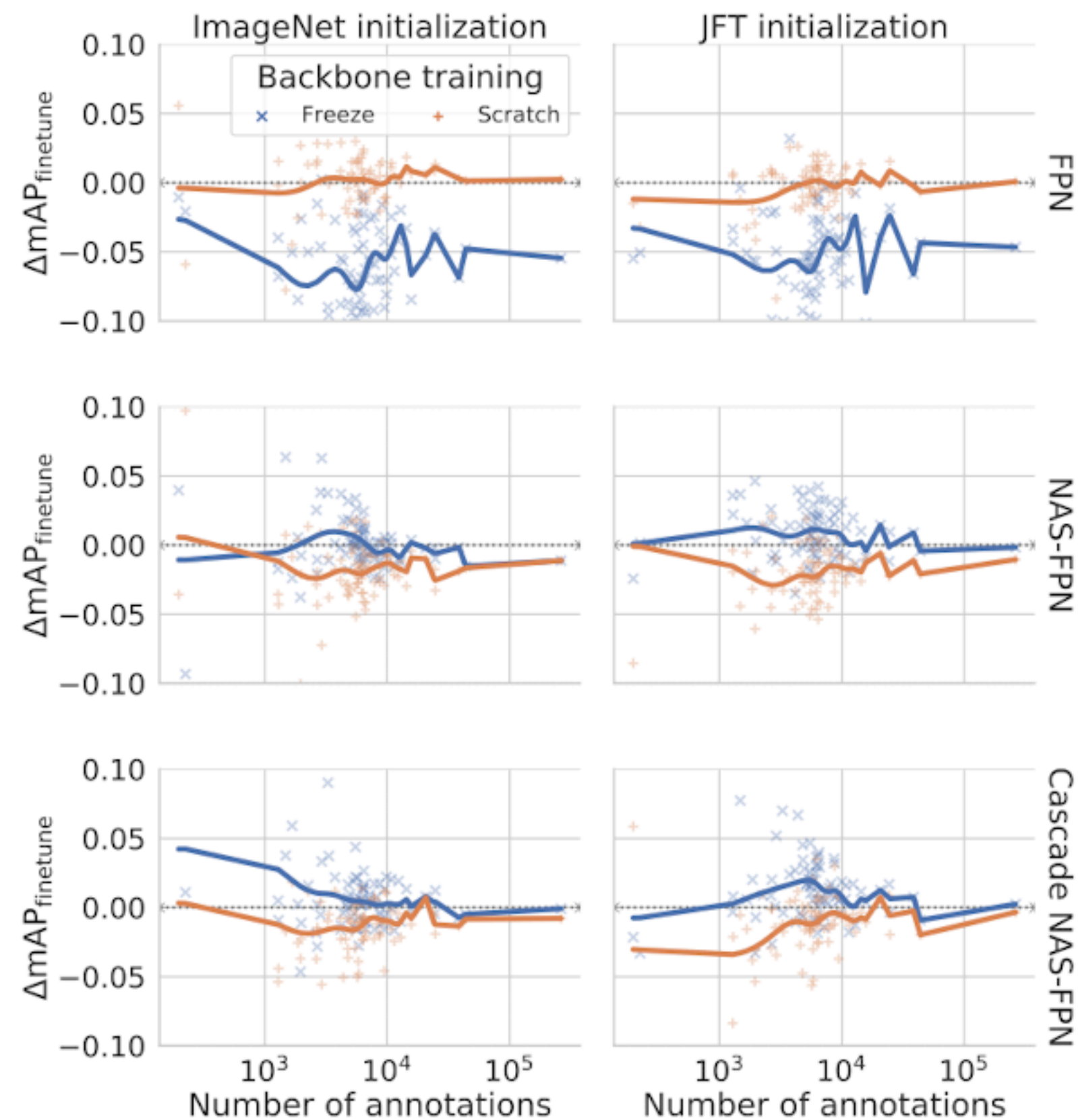


Figure 4. Class-wise mAP as a function of the number of training annotations *relative to fine-tuning the backbone*. Smoothed curves are obtained by applying Gaussian smoothing with $\sigma = 1000$. The ResNet-101 backbone is combined with detectors of varying capacities (rows) and initializations (columns). Freezing the backbone is increasingly beneficial as the detector capacity increases (top to bottom rows) and as the number of annotations decreases.

- 학습 데이터량에 따른 성능변화
 - FPN: Freeze의 경우 성능 감소
 - Cascade: data 개수가 적은 경우 상대적으로 큰 성능향상

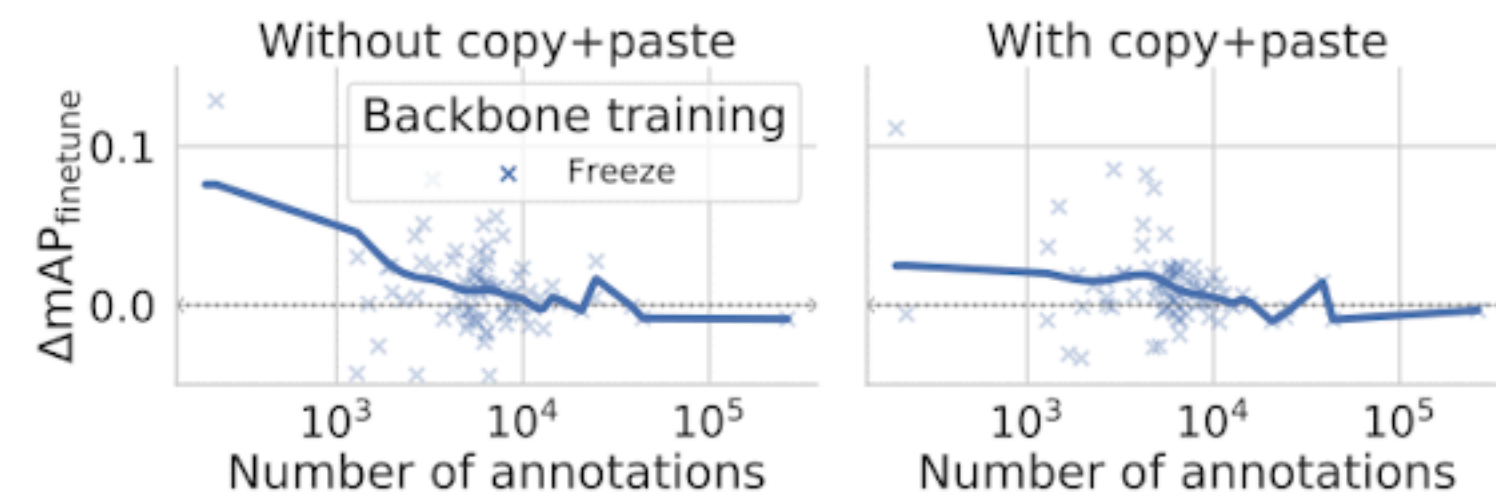


Figure 5. Class-wise mAP as a function of the number of training annotations *relative to fine-tuning the backbone*. Smoothed curves are obtained by applying Gaussian smoothing with $\sigma = 1000$. The EfficientNet-B7 architecture is trained without (left) and with (right) copy+paste augmentations.

- 데이터 수가 적은 경우 Freeze의 성능이 Fine-tune에 비해서 더 좋음

Model	Pretraining	mAP	AP @ 50
ResNet-101 + NAS-FPN + Cascade	ImageNet	51.1	69.1
	+ Freeze backbone	(+0.8) 51.8	(+0.8) 69.9
	+ Res. adapters	(+1.9) 53.0	(+2.4) 71.5
	JFT-300M	51.1	69.0
	+ Freeze backbone	(+1.7) 52.8	(+2.1) 71.1
	+ Res. adapters	(+2.5) 53.6	(+3.0) 72.0

Table 7. Adding residual adapters to the frozen backbone and training them along with the subsequent detector components yields another performance increase.

- Freeze backbone이 최선의 방안이 아님
 - Freeze backbone에 추가로 residual adapter를 추가한 경우 추가 성능 향상

Thanks